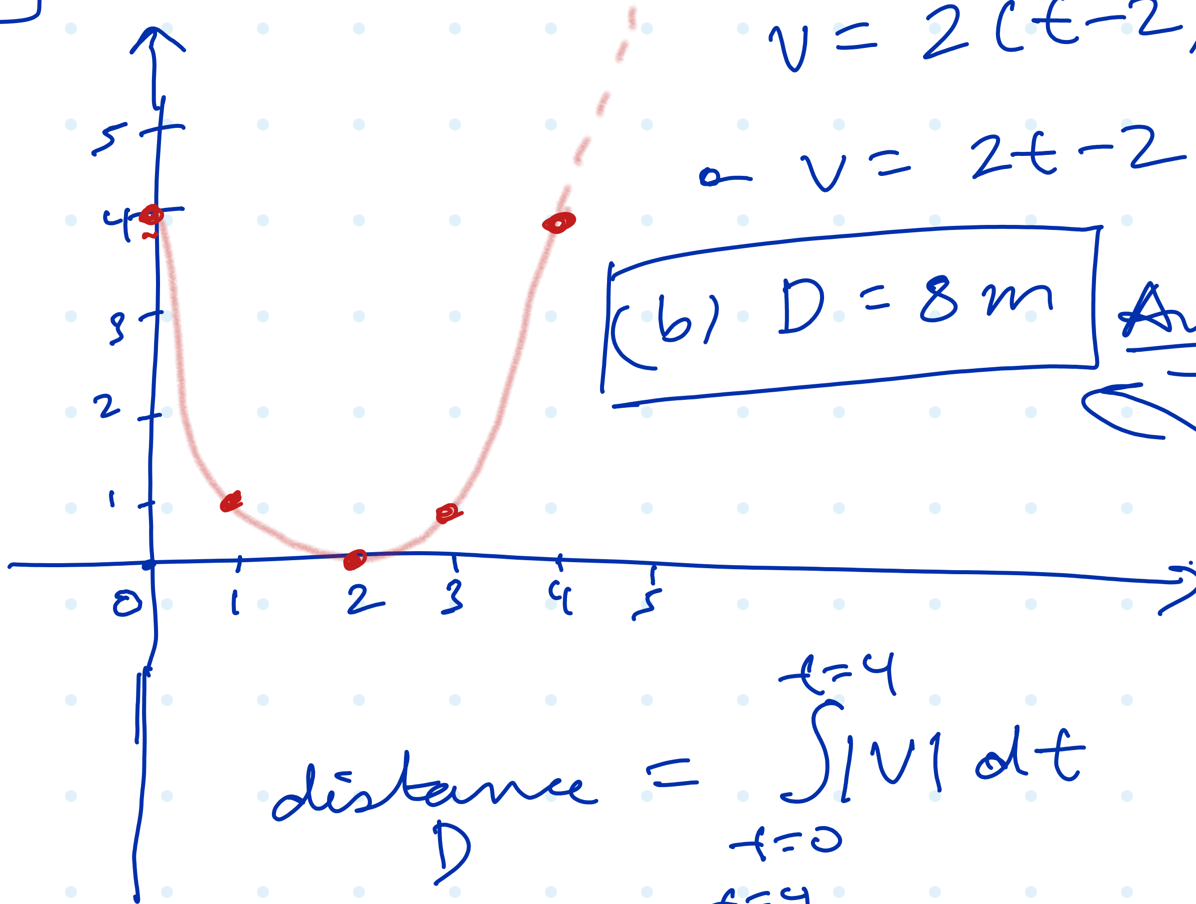
$\boxed{3.5}$ $x = (t-2)^2$



$v = 2(t-2)\cdot 1$

$\therefore v = 2t-2$

$\boxed{(b)\ D = 8\,m}$ Ans

distance $D = \int_{t=0}^{t=4} |v|\,dt$

distance $D = \int_{t=0}^{t=4} |2t-4|\,dt$

$t=0$ to $t=2$ $\qquad$ $t=2$ to $t=4$

$\therefore D = \int_{t=0}^{t=2} (4-2t)\,dt + \int_{t=2}^{t=4} (2t-4)\,dt$

$\therefore D = (4t-t^2)\Big|_{t=0}^{t=2} + (t^2-4t)\Big|_{t=2}^{t=4}$

$\therefore D = 4(2-0) - (4-0) + (16-4) - 4(4-2)$

$\therefore D = \cancel{8} - 4 + 12 - \cancel{8} = 8\,m$